

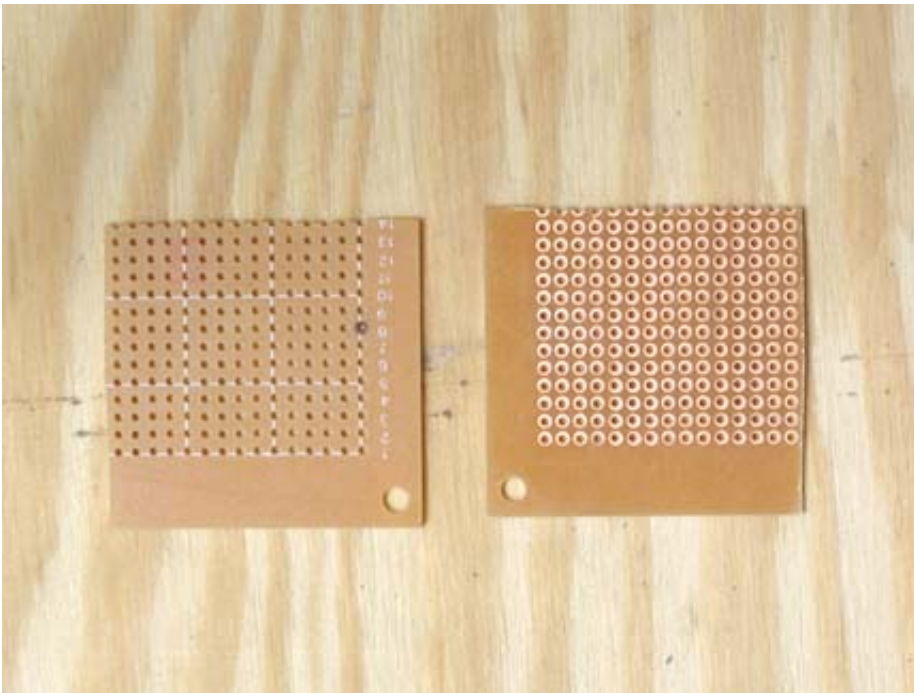
TP PHYSICAL COMPUTING

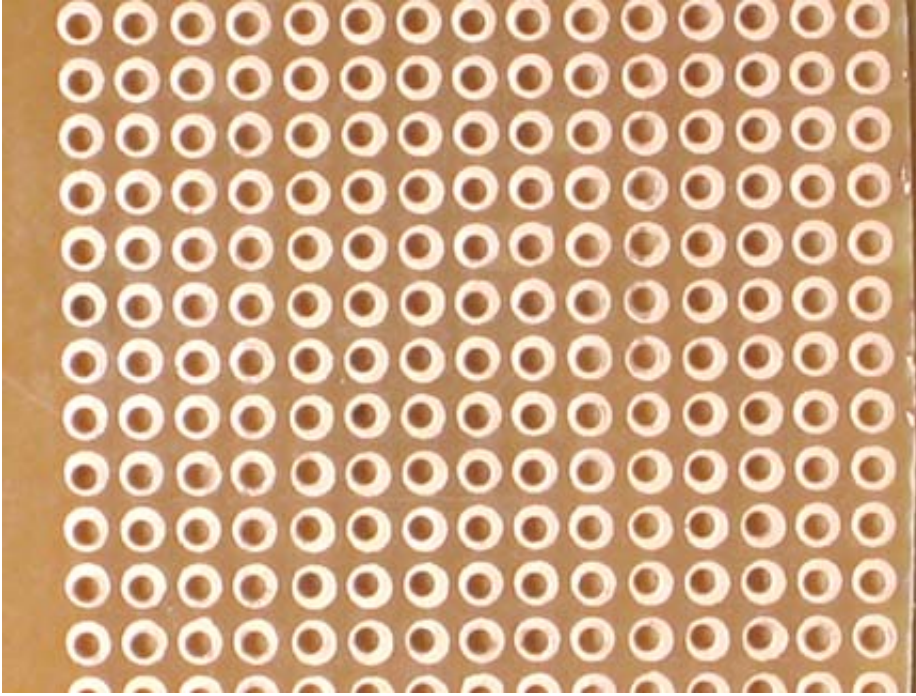
Soldering A Perf Board

This is a simple tutorial on soldering components to a perf board. The perf board is a great step between testing a circuit on a breadboard and making your own printed circuit board. The practice of building a circuit on a perf board gives you a level of robustness that the breadboard doesn't allow and gives you the flexibility to make changes that the PCB is incapable of.

A perf board is a heat resistant piece of glass-epoxy material with holes or perforations arranged in a grid. They are spaced at .1" or 2.5mm which makes them useful for a majority of components.

This is a piece of perfboard:



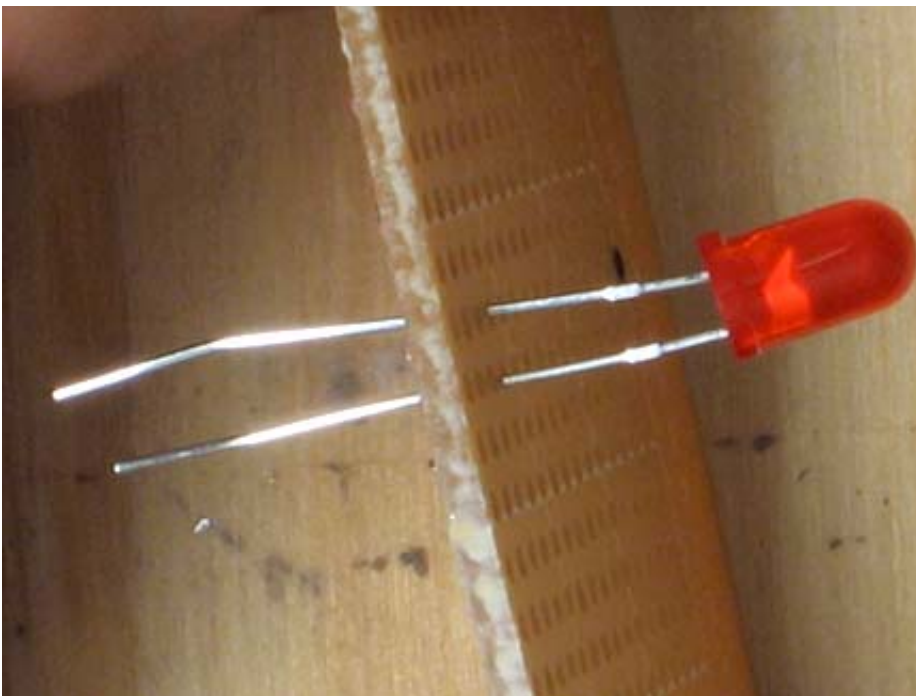


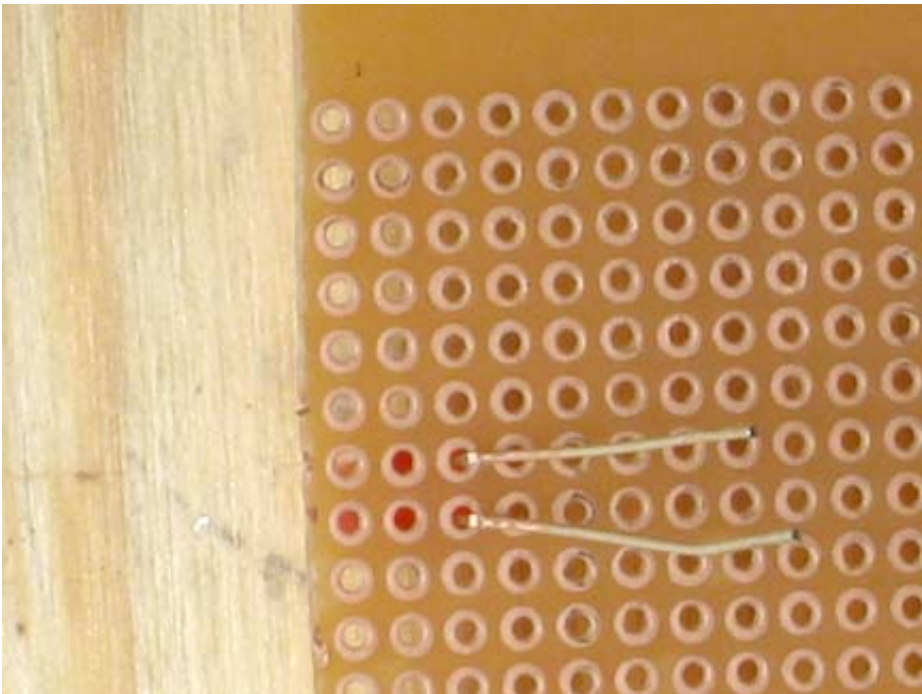
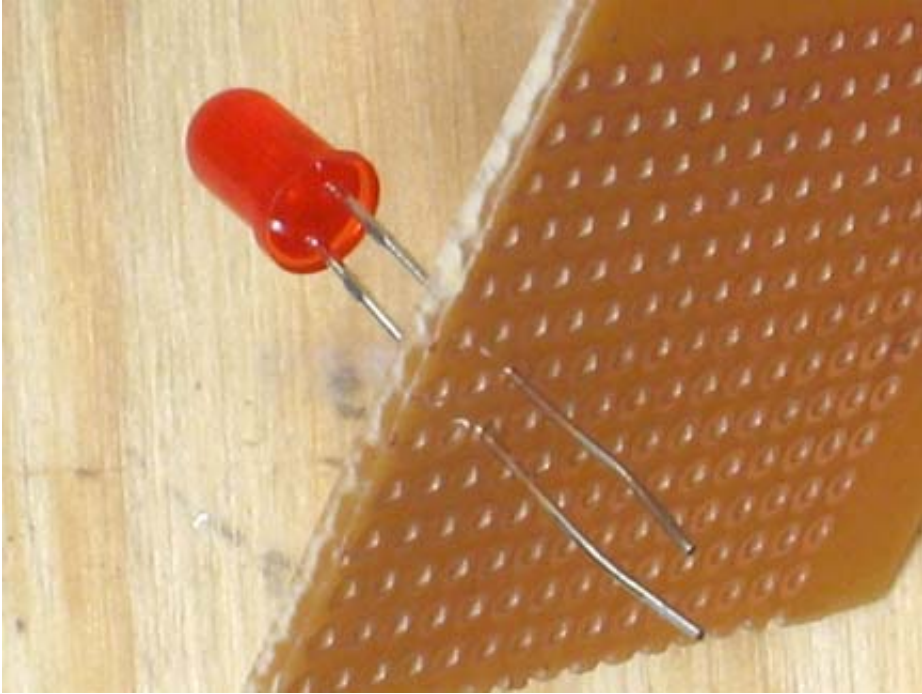
Notice how on one side of the board there are copper rings around the holes. This is the bottom of the board and the copper rings are called "pads" or "doughnuts".

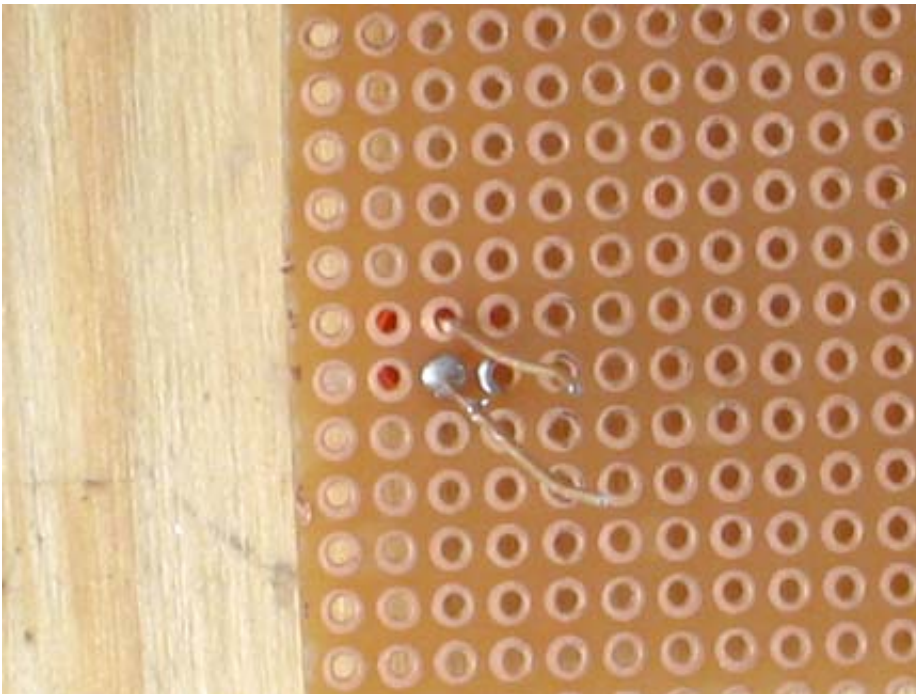
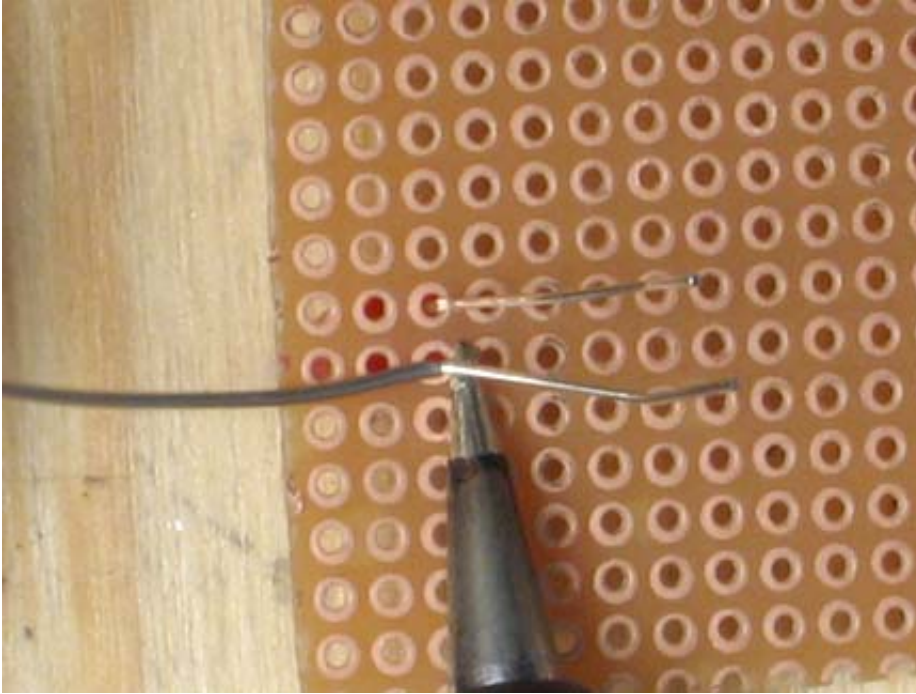
components are placed through the top of the board and soldered to the pads on the bottom.

This example is an LED and a resistor.

First we insert the LED.

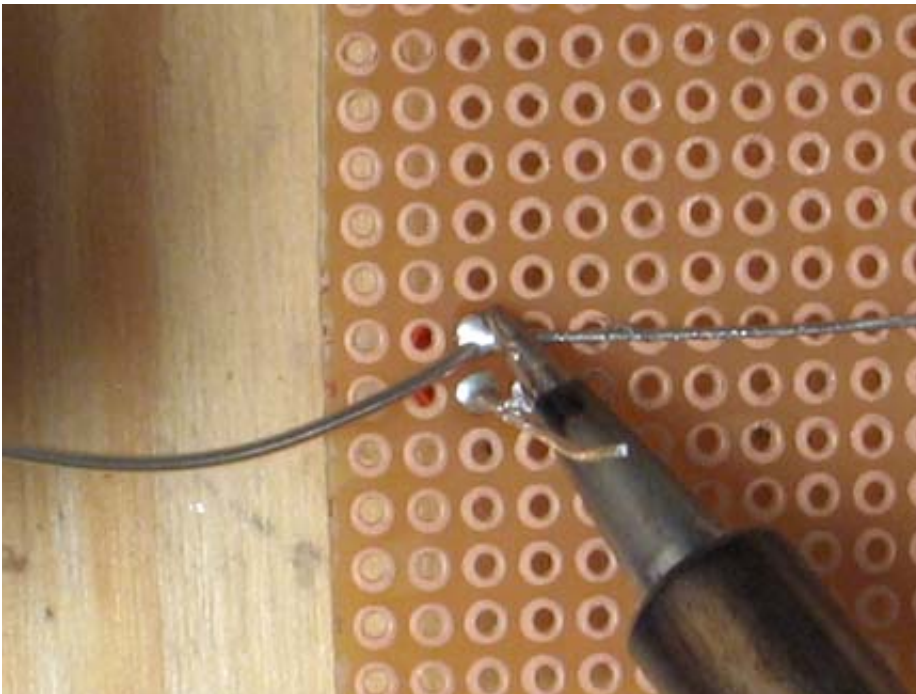
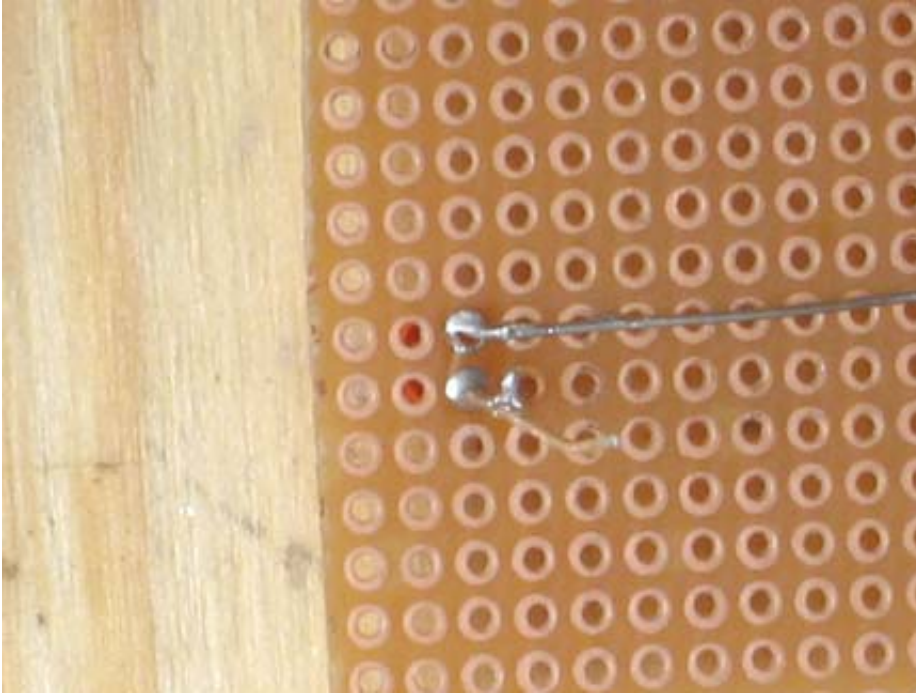






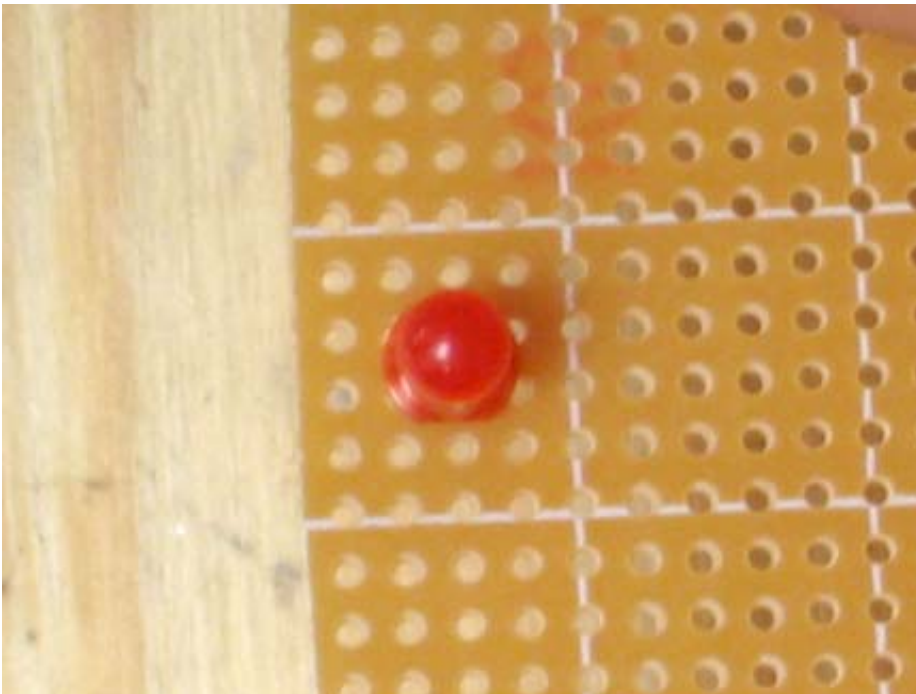
Holding the soldering iron to the pad around the leg of the component we add a little solder. With a good soldering iron at 350 degrees, a count of 5 is enough time to wait before adding the solder. When the copper of the pad is hot enough, the solder should melt and adhere itself to the pad and the component.

when it cools, the LED is held in place.

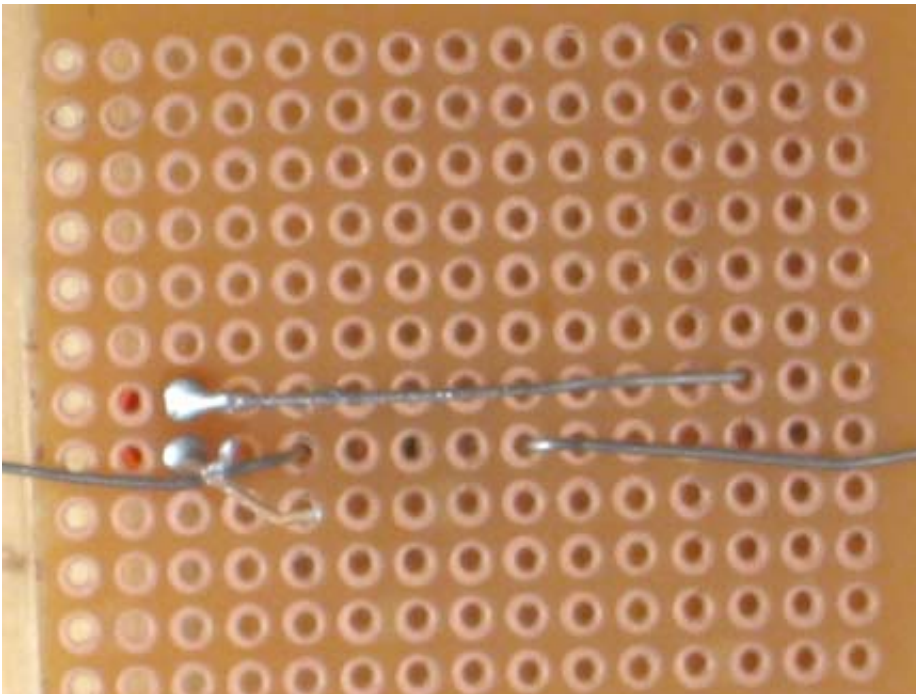
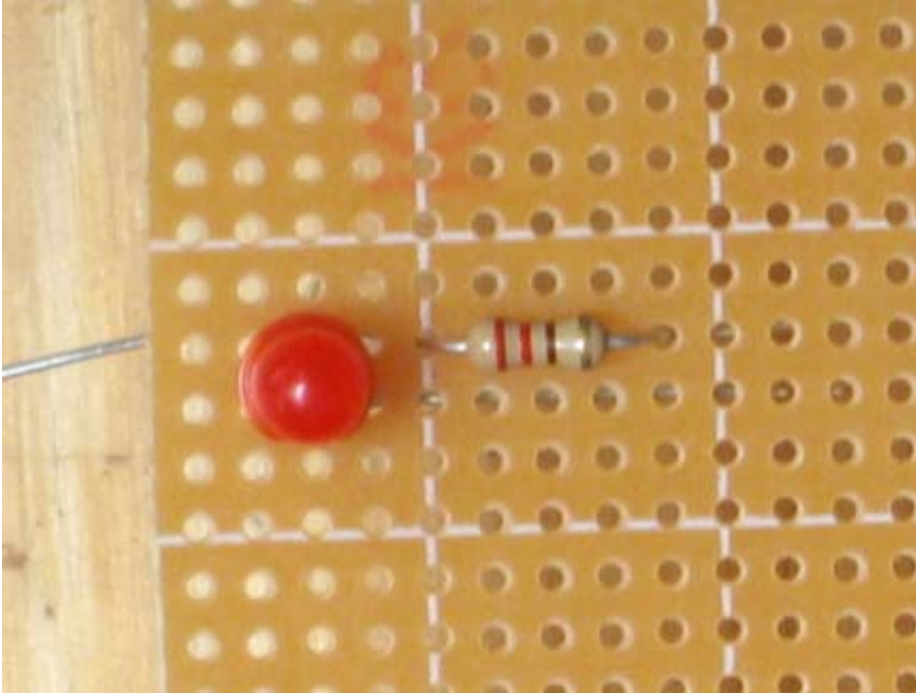


In this next step, look at how I am using the leg of the LED as a trace. A trace is a connection between two points on a board. Many times people making their first perfboard will try to connect the pads with a line of solder. This is a mess waiting to happen. The function of the pads when solder is applied to them is to pull the solder towards the component and away from the other pads.

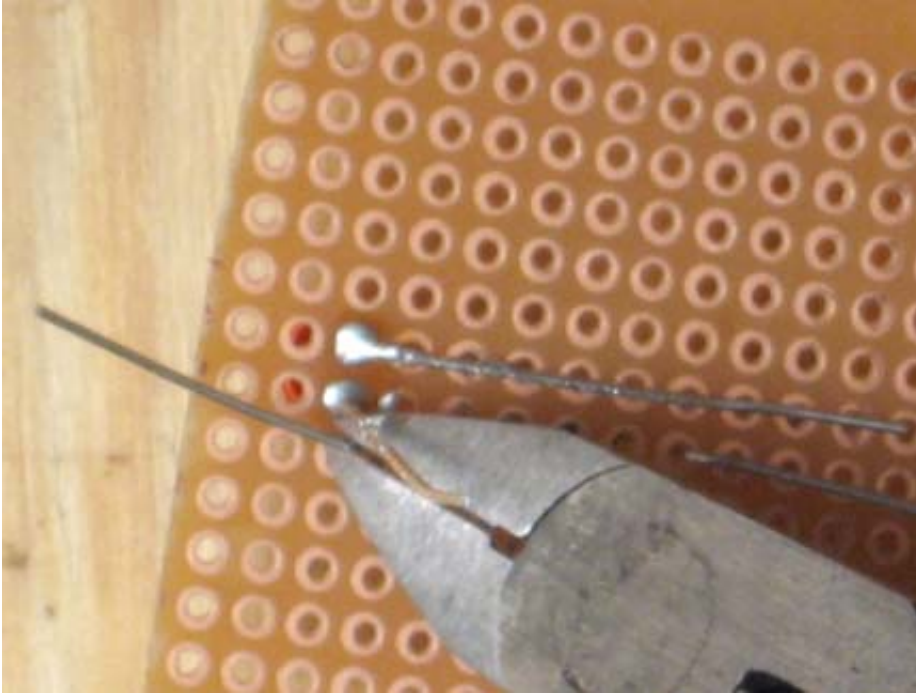
If you need a trace, using the leg of a component, a piece of wire, or a jumper are suitable methods. In any perfboard circuit you will find yourself using any three of these.



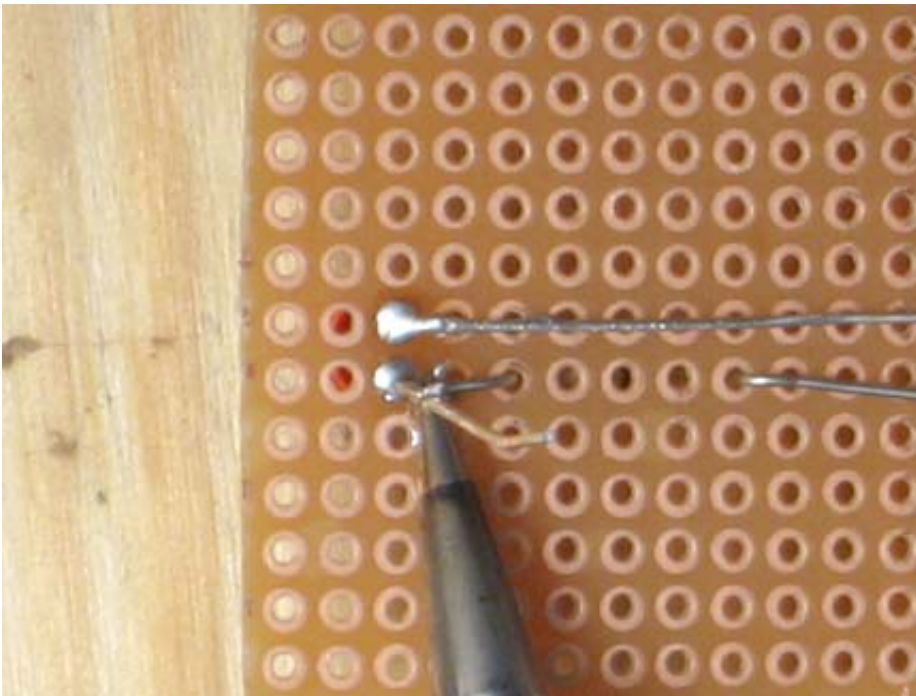
A quick note about soldering to a perfboard - I try to use solder as efficiently as possible. Too much solder and the board becomes a disaster. Lines become crossed and shorts develop. These can be difficult to find and correct. Too little solder and the connections are not robust. However, it's easier to add solder than to take away. Here I add a 220 resistor which can also work as a convenient jumper.



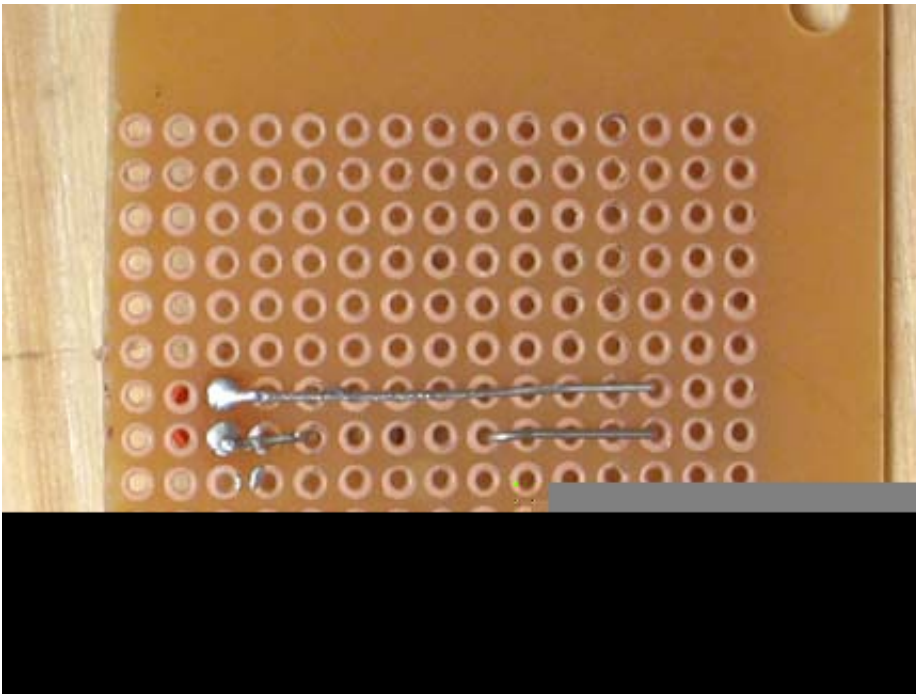
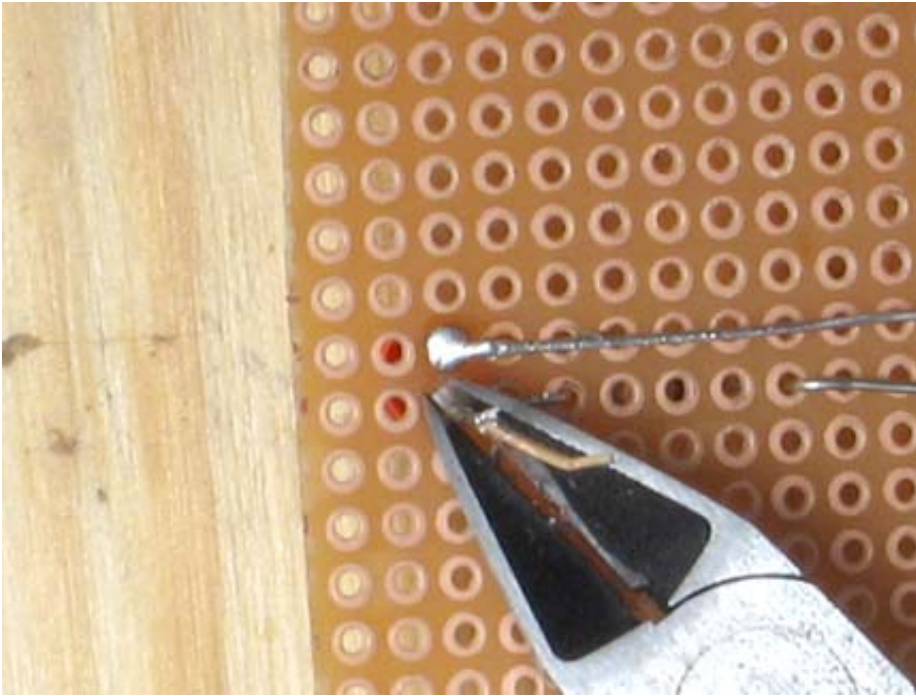
Trim the leg of the resistor to fit snugly against the leg of the LED.



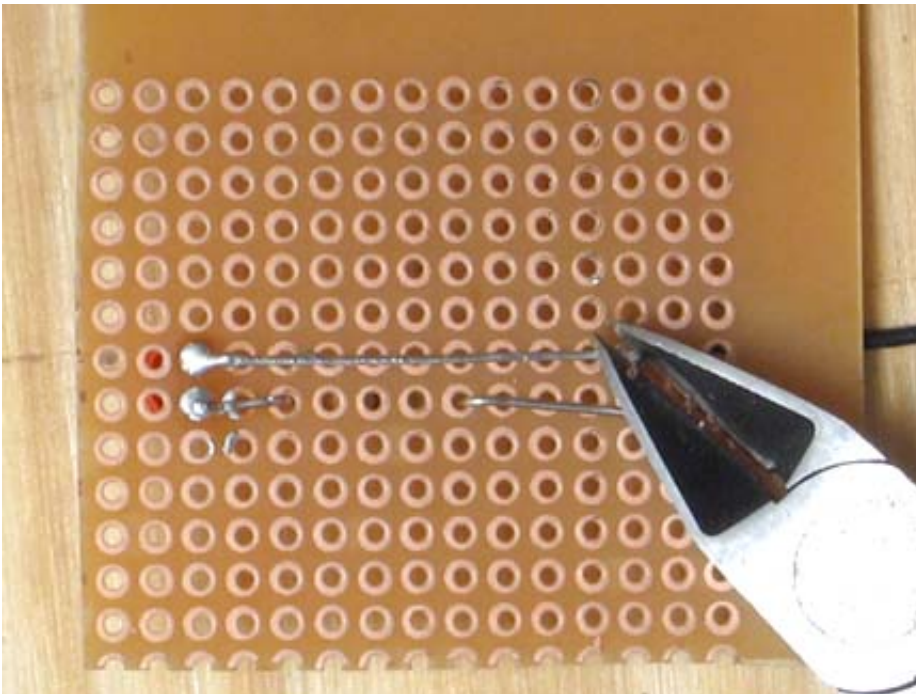
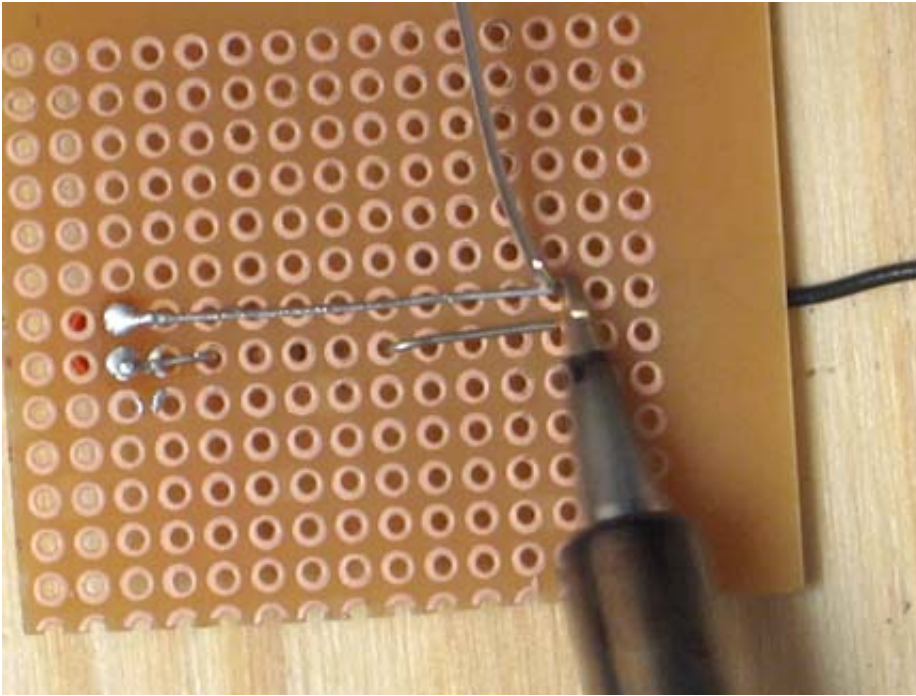
Here I am using the solder that is already holding the leg of the LED in place to connect the resistor. This is an efficient method of soldering part and keeps the underside neat.

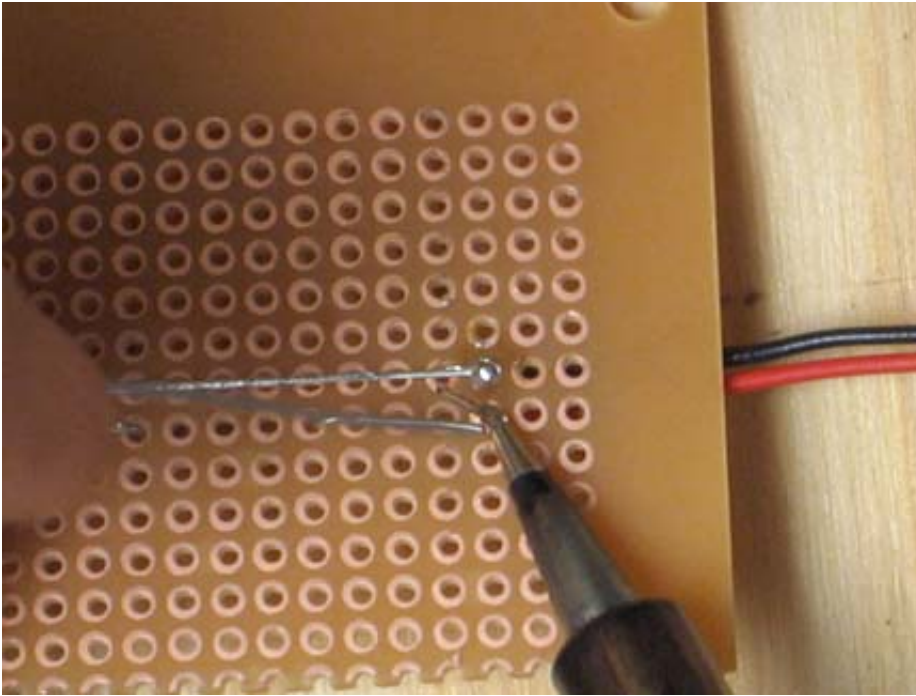


Snip the extra.

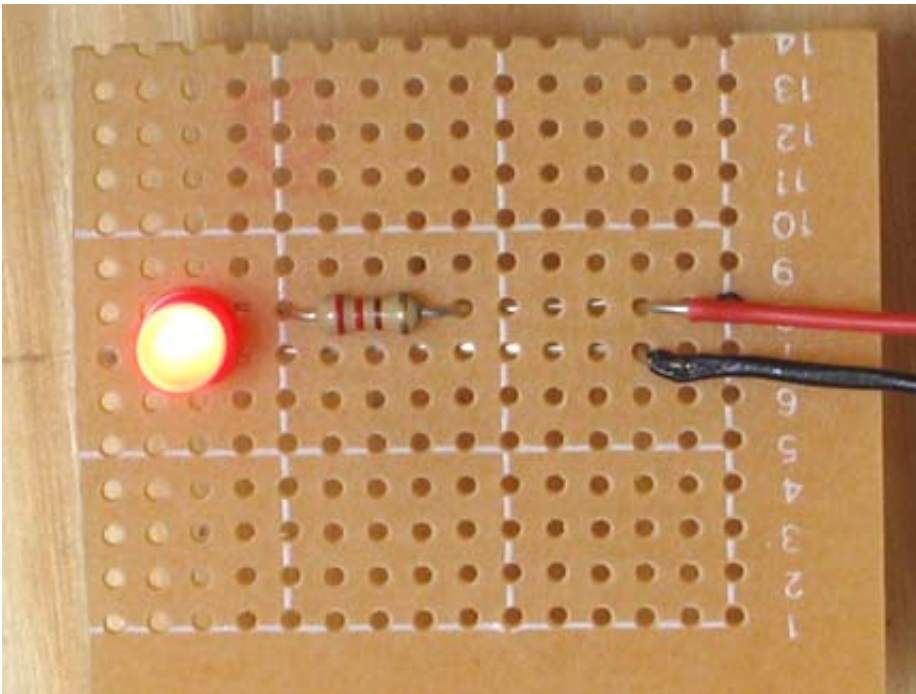


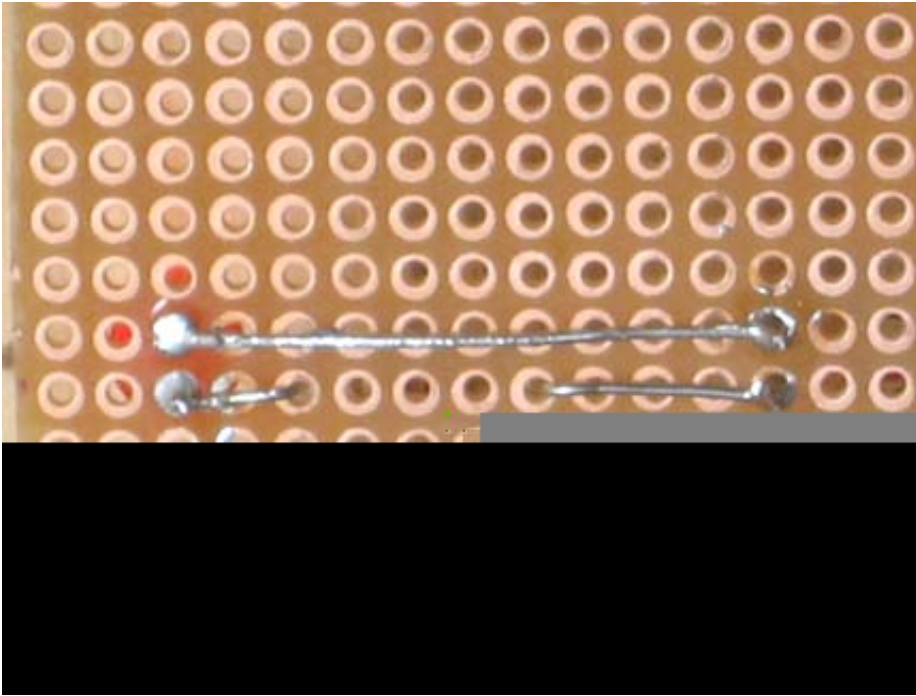
Here I add a wire for both power and ground.



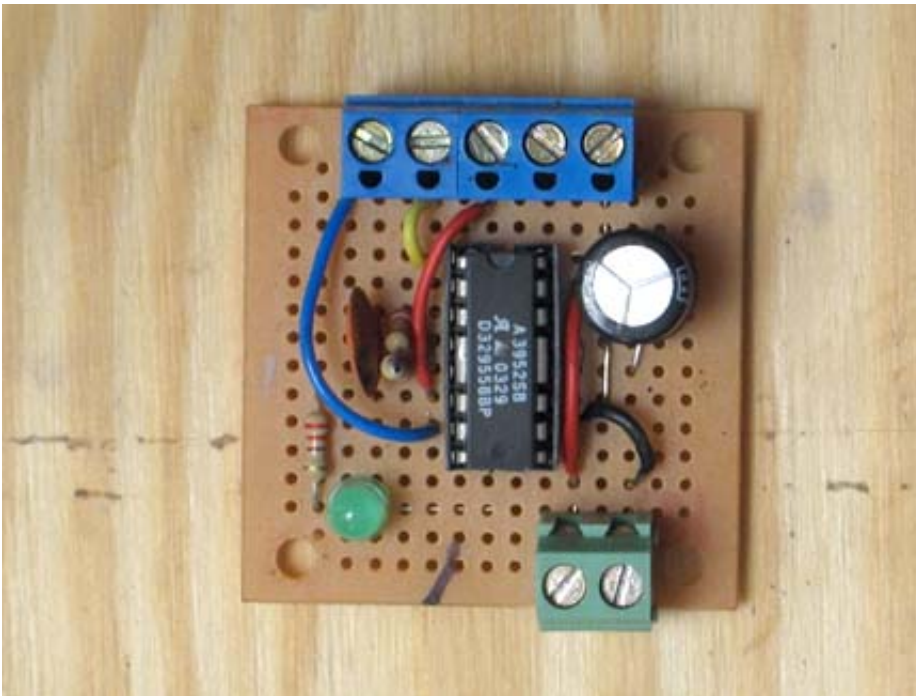


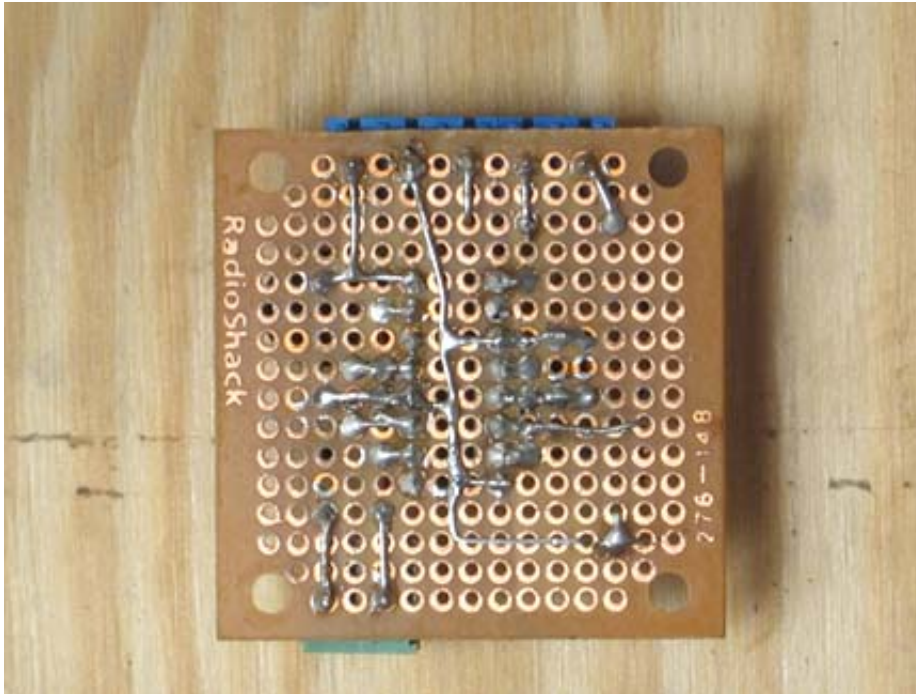
With the addition of a power and ground we can give 5v to the light.



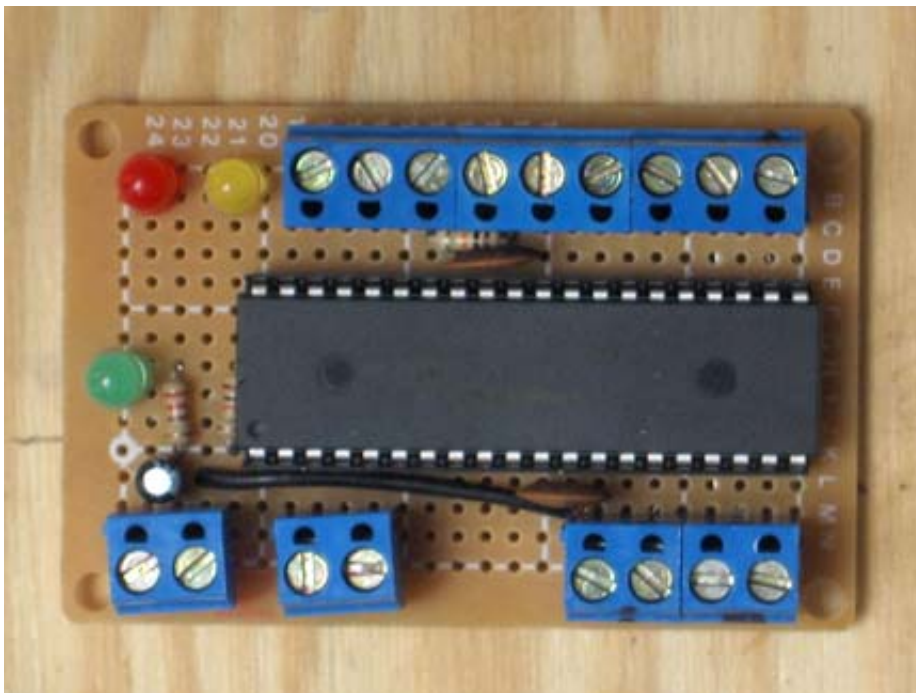


This is a very simple demo of soldering to a perfboard. Following are a few examples of other boards I have made. The first is a motor driver for the A3953 PWM motor driver and a board for any IC that can produce a PWM signal.





This is a board for a 40 pin PIC



Page last modified on August 16, 2007, at 10:27 AM